Higher National Diploma in Information Technology
First Year, Second Semester Examination – 2023
HNDIT2032- System Analysis & Design

Instructions for Candidates:
Answer any five (5) Questions
All questions carry equal marks.
Non-programmable calculators are allowed in this examination

No. of questions : 06
No. of pages : 04
Time : 03 hours

Question 01 [Total 20 Marks]

i. Give two (02) examples for operational stakeholders and two (02) examples for executive stakeholders in a Banking system. [04 Marks]

ii. List four (04) types of information systems based on the objective of the Information systems. [04 Marks]

iii. What is the role of SDLC in system development? Explain, Agile development methodology [06 Marks]

iv. Discuss the importance of a System Analyst possessing strong non-technical skills (e.g., communication, interpersonal skills) in addition to technical knowledge. Provide at least two specific examples where non-technical skills are critical for a System Analyst's success [06 Marks]

Question 02 [Total 20 Marks]

i. State two (02) system analysis activities. [04 Marks]

ii. What is *Feasibility Study* in a software development project? [04 Marks]

iii. Differentiate between *Payback Period* and *Net Present Value* as Project Evaluation Methods. List decision rules of each methods. [04 Marks]

iv. A company is considering investing in two projects. Details of the investments are as follows.

Project A: Initial investment of Rs. 600,000. Expected annual cash inflows of Rs. 180,000 for the next 5 years.

Project B: Initial investment of Rs. 450,000. Expected cash inflows: Year 1: Rs. 60,000, Year 2: Rs. 120,000, Year 3: Rs. 150,000, Year 4: Rs. 200,000, Year 5: Rs. 250,000.

a. Calculate the Net Present Value (NPV) of Project A and Project B, assuming a discount rate of 10%. [06 Marks]

b. Based on the NPV results, which project would you recommend the company undertake and why? [02 Marks]

Question 03 **[Total 20 Marks]**

i. Define a "*Business Process*" and explain its significance in the context of system analysis. [04 Marks]

ii. Explain the difference between *functional* and *non-functional* requirements in a system requirements specification, providing examples for each. [04 Marks]

iii. What is the role of *Use Case* diagrams in system Analysis? List one (01) advantage and one (01) limitation in the Use Case diagram. [04 Marks]

iv. Draw a Use Case diagram for an *Online Course Enrollment System* based on the following description:

Students can browse courses, enroll in courses, and view their enrolled courses. To enroll in a course, a student must first log in. Instructors can manage their courses (add/remove content, grade assignments). Administrators can manage student accounts (create, update, delete) and manage course catalogs (add/edit/delete courses). The system sends enrollment confirmations to students after successful enrollment.

[08 Marks]

Question 04 **[Total 20 Marks]**

i. Differentiate between *Context diagram* and *Level 0 Data Flow Diagram (DFD)*.

[04 Marks]

ii. List two (02) illegal data flows in DFD diagrams. [02 Marks]

iii. Answer the question based on following scenario. Scenario explains the process of a *Vehicle Rental System*

Customers can register their details, search for available vehicles, book a vehicle, and make payments. The system sends booking confirmations and rental agreements to customers. Rental Agents can add new vehicles to the fleet, update vehicle status (e.g., available, rented, under maintenance), and process vehicle returns. The system

provides vehicle availability reports to agents. Management can view overall rental statistics and generate financial reports. The system interacts with a Payment Gateway for online payments.

a. Identify the external entities of the *Vehicle Rental System.* (02 Marks)

b. Draw the Context Diagram (04 Marks)

c. Identify sub process of the system (02 Marks)

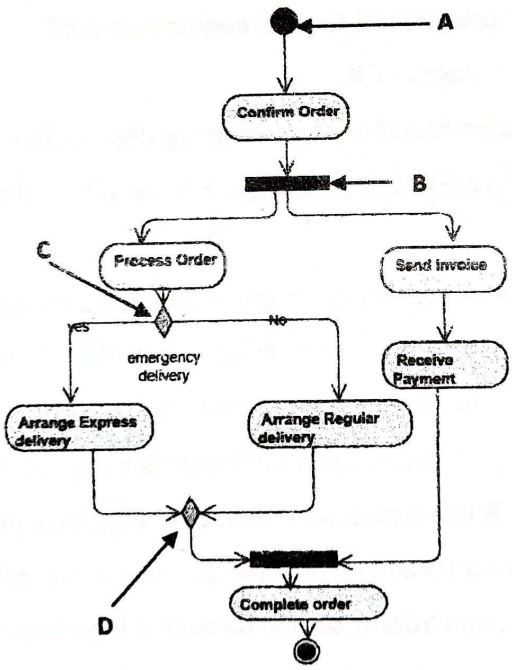
d. Identify the data stores of the system. (02 Marks)

e. Draw the Level-0 DFD for the above scenario. (04 Marks)

Question 05 **[Total 20 Marks]**

i. What is *Activity Diagram*? List two (02) advantages of creating activity diagram. [04 Marks]

ii. Name the following symbols (A → D) of the activity diagram. [04 Marks]



iii. Give examples for following terms related to Entity-Relationship Diagram (ERD).

a. Primary key

b. Weak entity

c. Associative Entity

d. Multi-valued attribute [04 Marks]

iv. Draw an Entity-Relationship Diagram (ERD) based on the following business rules for a University Management System. Add at least two attributes appropriately for each entity, including the Primary key.

A University has many Departments. Each Department belongs to only one University. A Department offers many Courses. Each Course is offered by only one Department. A Course can have many Students enrolled. A Student can enroll in many Courses. A Professor teaches many Courses. A Course is taught by only one Professor.

[08 marks]

Question 06 **[Total 20 Marks]**

i. Define unit testing and state two (02) limitations of it. [04 Marks]

ii. Differentiate between *Black-Box Testing* and *White-Box Testing* as testing approaches. [04 Marks]

iii. *Parallel Running* is one (01) of the file conversion methods. List two (02) advantages and two (02) disadvantages of it. [04 Marks]

iv. Draw the class diagram based on the following description:

The Online Bookstore System is designed to facilitate the Browse and purchase of books by customers.

Each book within the inventory is uniquely identified by an ISBN, and has a specific title, a designated author, and a set price. An author is known by their name. It's common for a single author to have written multiple books. Customers interact with the system to make purchases. Each customer is assigned a unique customer ID and has personal details. When a customer decides to buy, they place an Order. Every order has its own unique order ID and records the date it was placed, along with the calculated total amount. Finally, an Order is settled through a Payment. Each payment records the amount, the date it was made, and the payment method used (e.g., credit card, PayPal).

[08 Marks]